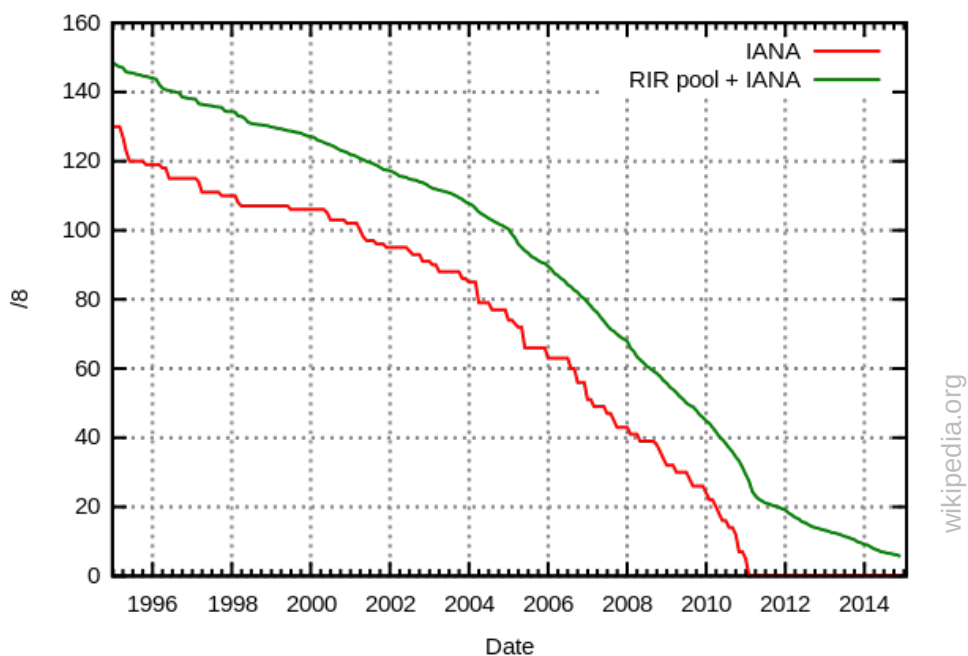


Internet Protocol – version 6 IPv6

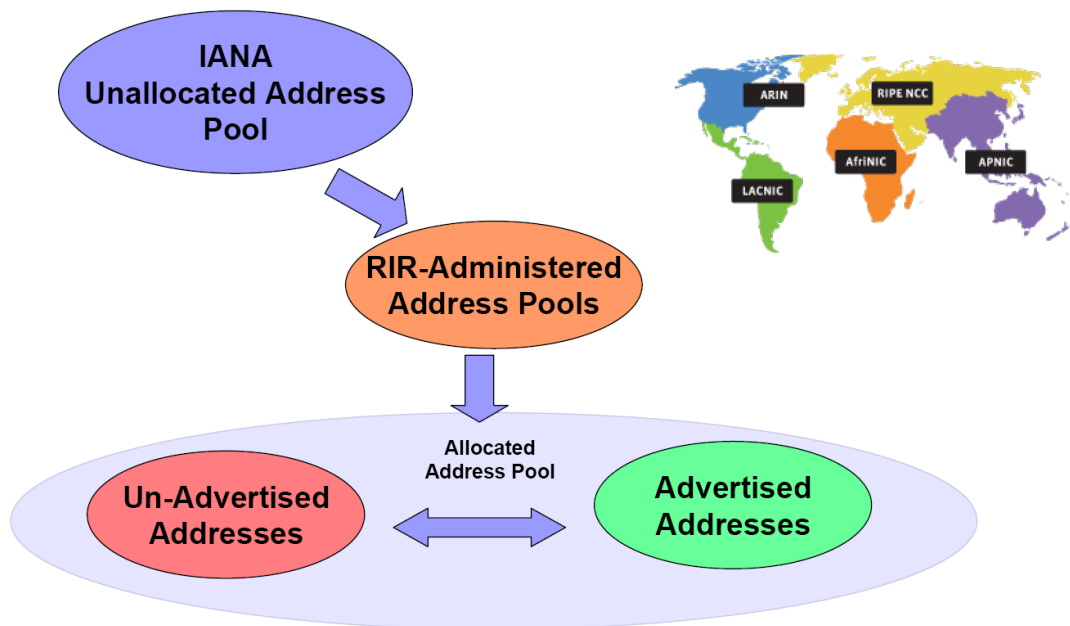
Histórico

v3.3, Miguel Leitão, 2024

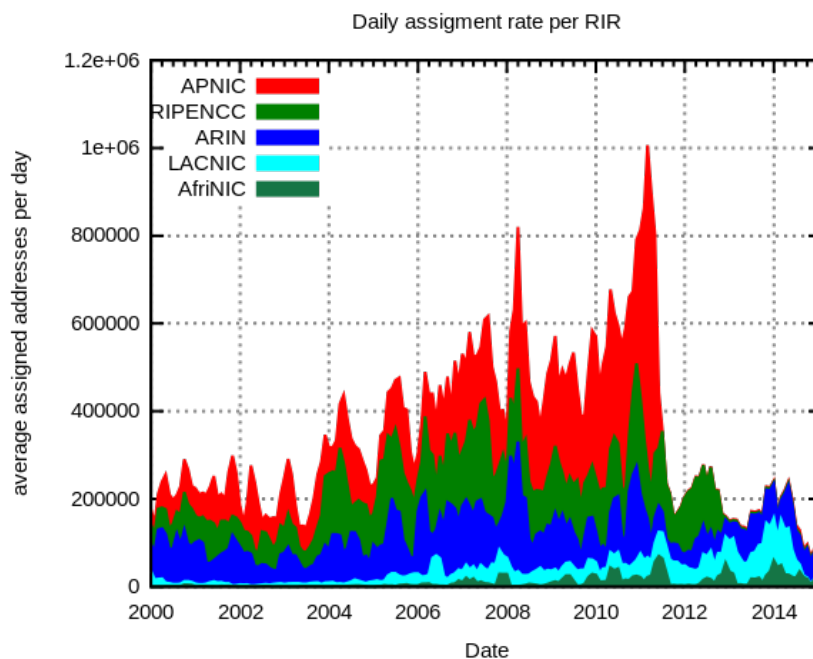
Endereços IPv4 disponíveis



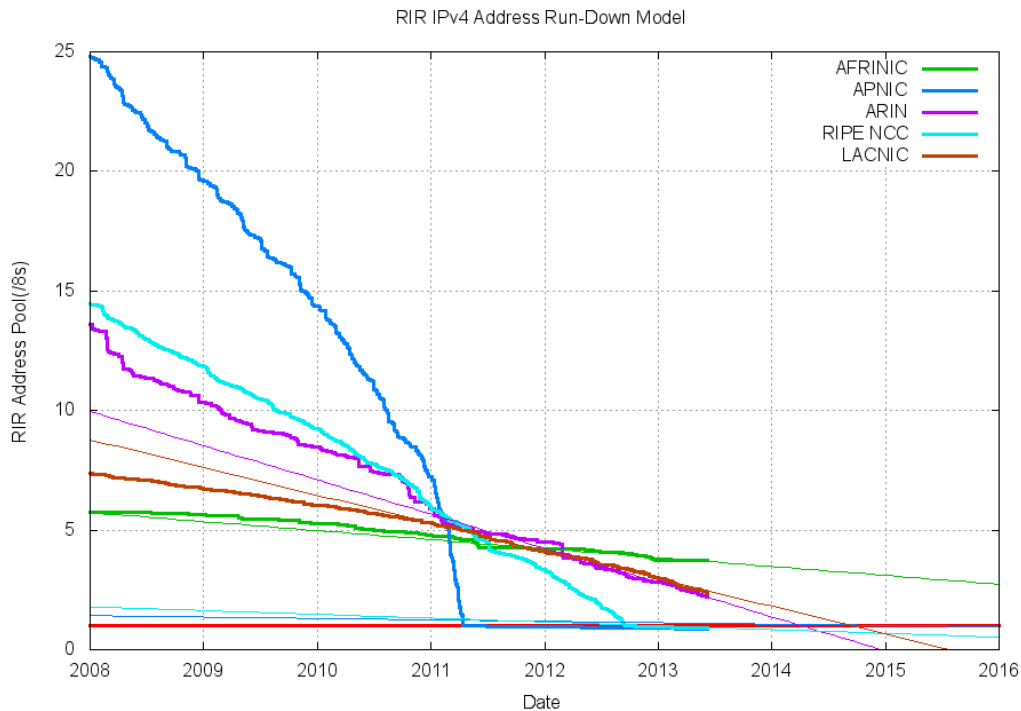
Alocação de endereços



Consumo de Endereços IPv4



Previsão



IPv6 - Internet Protocol Version 6

- ➔ Problemas com a actual versão actual do Protocolo IP (IPv4)
 - Número insuficiente de endereços
 - Espaço de endereçamento quase esgotado
 - Pouca flexibilidade no tipo de serviços suportados
 - Tamanho das tabelas de encaminhamento dos routers próximo de um limite insustentável ao nível capacidade/rapidez
- ➔ Objectivos da nova versão do Protocolo IP (IPng ou IPv6)
 - Suportar um número elevado de máquinas
 - Reduzir o tamanho das tabelas de encaminhamento
 - Simplificar o protocolo de forma a tornar os routers mais rápidos
 - Oferecer melhor segurança
 - Melhor suporte de vários tipos de serviços (tempo real)
 - Suportar máquinas móveis sem alteração de endereço
 - Garantir a co-existência de várias versões IP
 - Desenvolver um protocolo flexível que permita evolução

Endereço IPv6

128 bits

Address space:

340,282,366,920,938,463,374,607,431,768,211,456 endereços

3.4×10^{38} endereços

$6,65 \times 10^{23}$ endereços/m²

- Representa-se em Hexadecimal,
- em grupos de 16 bits separados por ':'
 - 2001:0DF8:0101:0000:0000:00E0:F796:4F31
- Zeros à esquerda de cada grupo podem ser omitidos
 - 2001:DF8:101:0:0:E0:F796:4F31
- :: Substitui grupos em falta por zeros
 - 2001:DF8:101::E0:F796:4F31

Exemplos de endereços IPv6

- Global unicast address is:
2001:DF8:101:1::E0:F796:4F31,
subnet is 2001:DF8:101:1::0/64
- Link-local address is FE80::80:9341:A892
- Unspecified Address is 0:0:0:0:0:0:0:0 or ::
- Loopback Address is 0:0:0:0:0:0:0:1 or ::1
- Group Addresses (Multicast)
 - FF02::9 for RIPv6

IPv6 - Endereçamento

→ Prefixo

- FEDC:BA98:76::1234:5678:9ABC / 64
 - os primeiros 64 bits representam o prefixo

→ Tipos de endereços

- Não existe o conceito de classes
- Não existem endereços de Broadcast
- Endereços Unicast
- Endereços Anycast
 - Identificam 1 grupo de interfaces de nós diferentes
 - Não pode ser usado como endereço de origem
 - Permite aceder mais rapidamente a um servidor ou serviço
 - Conjunto de servidores configurados com um endereço anycast
 - O host acede ao servidor mais próximo
- Endereços Multicast
 - Semelhante ao Anycast mas o pacote é recebido por todos os membros do grupo

IPv6 – Endereços Unicast

→ Global Unicast

- Endereços organizados de forma hierárquica para permitir diminuir as tabelas de encaminhamento
 - Prefixo de 3 bits (010) e 4 campos
 - Registry ID – Identifica entidade responsável pela gestão de espaço de endereçamento (prevendo que isto possa ser atribuído a várias organizações para além de IANA)
 - Provider ID – Identifica 1 provider
 - Subscriber ID – Identifica assinantes ligados a 1 provider
 - Infra-Subscriber ID

→ Loopback

- 0:0:0:0:0:0:0:1 ou ::1
- Equivalente ao 127.0.0.1 do IPv4

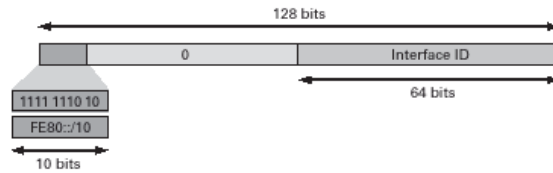
→ Não especificado

- 0:0:0:0:0:0:0:0 ou ::
- Ausência de endereço - utilizado pelas máquinas quando executam a auto-configuração
- Equivalente ao 0.0.0.0 do IPv4

IPv6 – Endereços Unicast

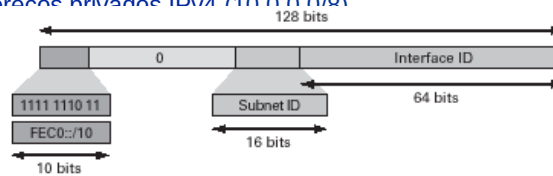
→ Link-local

- Utilizados apenas dentro de uma rede local: routers não encaminham estes pacotes
- Identificados por prefixo de 10 bits (1111111010)



→ Site-local

- Utilizados para endereçamento dentro de uma instituição
- Permitem criação de sub-redes
- Routers não encaminham estes pacotes para a Internet
- Equivalentes aos endereços privados IPv4 (10.0.0.0/8)

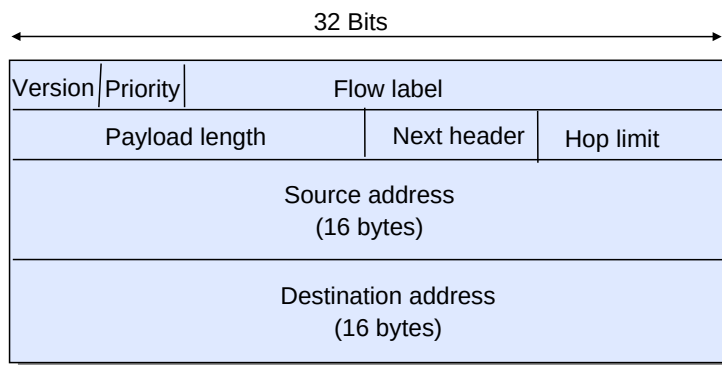


Well-Known Multicast Addresses

→ Link-local scope

- **FF02::1** All Nodes
- **FF02::2** All Routers
- **FF02::1:2** All DHCP Agents

IPv6 - Cabeçalho



- Endereços de 16 bytes ➡ resolve problema de escassez de endereços
- Cabeçalho mais simples ➡ aumenta a rapidez de processamento (40 bytes fixos)
- Cabeçalhos opcionais ➡ só processados pelas entidades adequadas

IPv6 - Cabeçalho

➔ Version

- Identifica qual a versão do protocolo (4 ou 6)
- Permite a co-existência de IPv4 e IPv6

➔ Priority

- 0 a 7 - para tráfego não prioritário (1:news; 4:FTP; 6:Telnet)
- 8 a 15 - para tráfego com exigências de tempo real (audio e vídeo)

➔ Flow label

- permite estabelecer pseudo-ligações entre uma origem e um destino caracterizada, por exemplo, pela existência de banda reservada

➔ Payload length

- Equivalente a *TotalLength* de IPv4 mas não inclui os 40 bytes de cabeçalho

➔ Next header

- Identifica qual dos cabeçalhos de extensão vem a seguir ou (se for o último cabeçalho IP) qual dos protocolos de transporte (TCP ou UDP) usado

➔ Hop limit

- Idêntico a TTL de IPv4

➔ Source/Destination address

- Endereços de Origem e Destino
- 16 bytes (128 bits)

IPv6 - Cabeçalhos de Extensão

→ Hop-by-Hop options

- informação vária para routers e que deverá ser analisada em todos os routers
p.e: envio de jumbograms (> 65 536)

→ Routing

- definição da rota total ou parcial a seguir

→ Fragmentation

- Fragmentação no IPv6 deve ser efectuada pelo host emissor e não pelos routers
→ necessário saber o MTU mínimo do caminho que vai ser percorrido
- Contém informação do tipo: Datagram identifier, Fragment number, more fragment bit

→ Authentication

- Mecanismo que permite assegurar ao receptor a veracidade da identificação do emissor

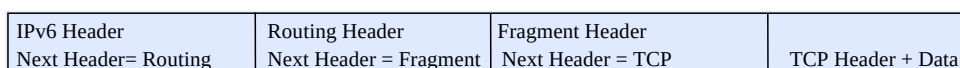
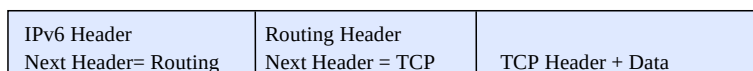
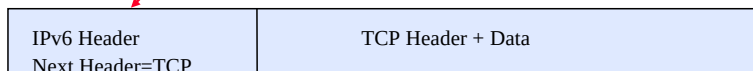
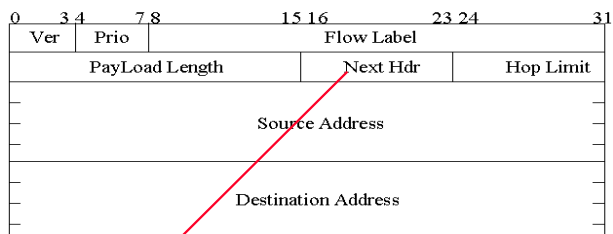
→ Encapsulation security payload

- Permite codificar o conteúdo de forma a que apenas o destinatário o consiga decodificar

→ Destination

- informação adicional para ser analisada pelo destinatário
 - opções a definir no futuro

IPv6 - Cabeçalhos de Extensão



Cabeçalhos IPv4 vs IPv6

- ➔ IPv4 **IHL** - não necessário porque tamanho do cabeçalho IPv6 é fixo
- ➔ IPv4 **Protocol** - *Next header* identifica o protocolo de transporte
- ➔ Campos de IPv4 relacionados com **fragmentos** desaparecem
 - IPv6 exige que todas as máquinas suportem pacotes de 576 bytes (fragmentação menos frequente)
 - Se um router recebe 1 pacote grande, não o fragmenta ➔ envia mensagem de erro a pedir pacotes mais pequenos

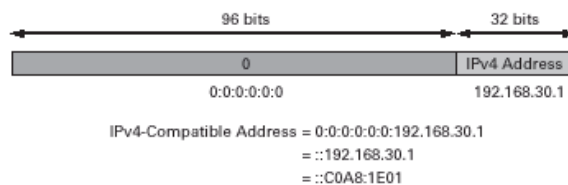
↓
Mais eficiente que fragmentação!
- ➔ IPv4 **Checksum** - tempo de cálculo elevado!

↓
Pouco eficiente!
Níveis 2 e 4 já têm esta funcionalidade!
- ➔ IPv4 address tem 32 bits e IPv6 address tem 128 bits
 - Se os endereços forem atribuídos ao ritmo de 1 milhão por microsegundo, demorará 20 anos a esgotar os endereços
- ➔ Serviços especiais são colocados em cabeçalhos de extensão
 - Alguns não necessitam de ser analisados pelos routers mas apenas no destino

IPv6 – IPv4

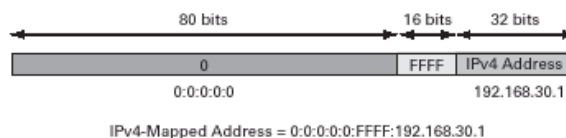
➔ IPv4 compatível

- Descontinuado



➔ IPv4 mapped

- Utilizado quando um host IPv6 precisa de comunicar com um host IPv4



Windows XP -2

```
C:\Documents and Settings\deepak>ipconfig
```

Windows IP Configuration

Ethernet adapter Local Area Connection:

```
Connection-specific DNS Suffix . :  
IP Address. . . . . : 172.31.2.60  
Subnet Mask . . . . . : 255.255.0.0  
IP Address. . . . . : 2001:e30:1401:2:d46e:b891:3082:b939  
IP Address. . . . . : fe80::202:b3ff:fe3e:12b6%6  
Default Gateway . . . . . : 172.31.1.250  
                             fe80::20d:65ff:fef9:7070%6
```

Linux

→ ifconfig

```
[root@mail sysconfig]# ifconfig
```

```
eth0      Link encap:Ethernet  HWaddr 00:12:79:95:D2:1B  
          inet addr:202.141.43.130  Bcast:202.141.43.255  Mask:255.255.255.128  
          inet6 addr: 2001:e30:1400:1:212:79ff:fe95:d21b/64 Scope:Global  
          inet6 addr: fe80::212:79ff:fe95:d21b/64 Scope:Link  
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1  
          RX packets:2088779 errors:0 dropped:0 overruns:0 frame:0  
          TX packets:123384 errors:0 dropped:0 overruns:0 carrier:0  
          collisions:0 txqueuelen:1000  
          RX bytes:1243359878 (1.1 GiB)  TX bytes:14476453 (13.8 MiB)  
          Interrupt:209
```

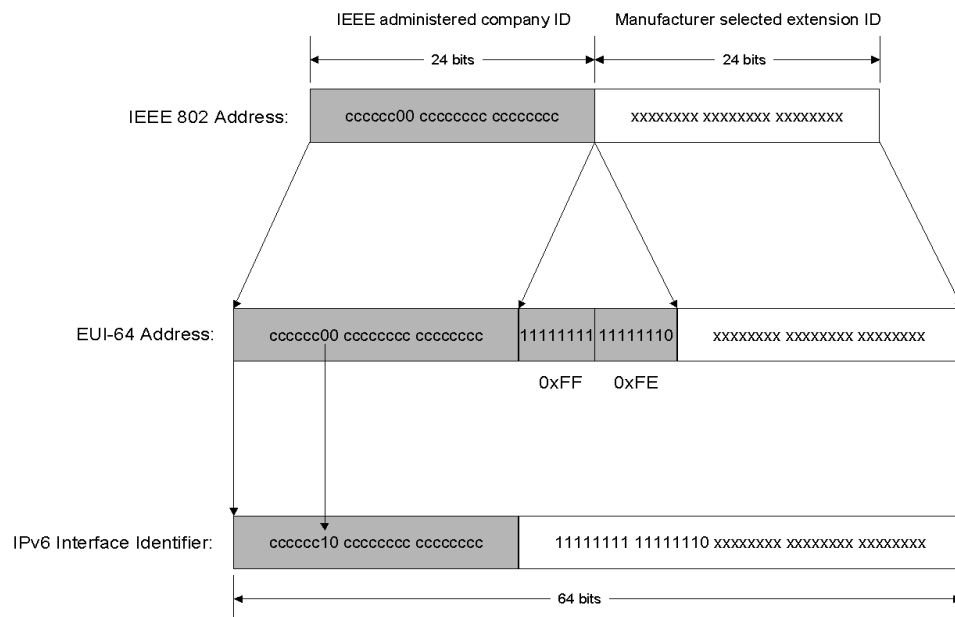
IPv6 Autoconfiguration

- Manual configuration of individual machines should not be required.
- Address autoconfiguration assumes that each interface can provide a unique identifier for that interface (i.e., an "interface token")
- Plug-and-play communication is achieved through the use of link-local addresses
- Small sites should not need stateful servers
- A large site with multiple networks and routers should not require the presence of a stateful address configuration server.
- Address configuration should facilitate the graceful renumbering of a site's machines

Configuração de endereço IPv6

1. Determinar identificador IPv6/EUI-64
2. Determinar endereço stateless link-local
3. Configurar e verificar endereço stateless link-local
4. [Solicitar informação sobre router.]
5. Escutar anúncios do router
6. Utilizar prefixo anunciado pelo router para configurar endereço stateless.
7. Se router indica a utilização de endereço stateful, usar DHCP para obter endereço.

IPv6 Interface Id



Link-local SLAAC

Link-Local Stateless Auto-Configured Address:

FE80 ::

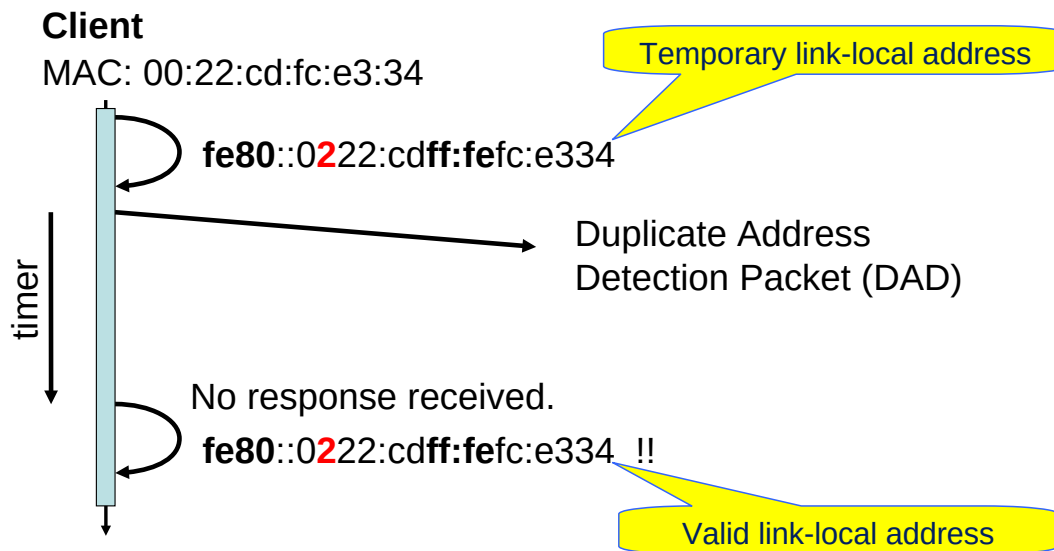
Intf ID

Exemplo:

- MAC: 00-AA-00-3F-2A-1C
- EUI-64: 00-AA-00-FF-FE-3F-2A-1C
- Intf ID: 02-AA-00-FF-FE-3F-2A-1C
- Notação Hexadecimal 2AA:FF:FE3F:2A1C
- Link-Local Address: FE80::2AA:FF:FE3F:2A1C

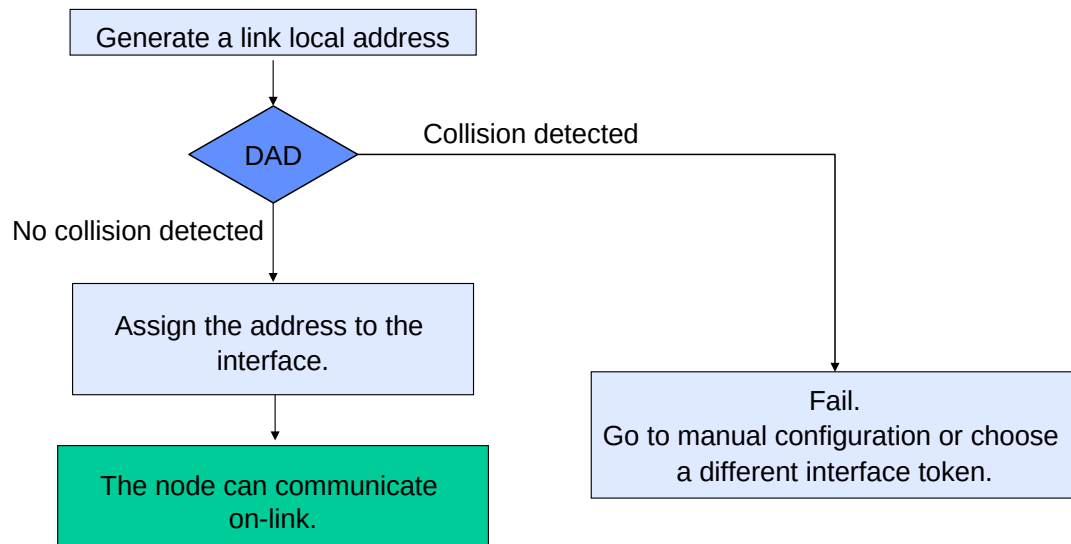
•Existem alternativas com variação temporal para protecção de identidade (RFC 3041).

Link Local Address DAD



This link local address can be used to communicate with other hosts on the same link.

Stateless Autoconfiguration



Stateless Autoconfig

- Routers should send out router advertisements (RA) at regular intervals to the all-hosts address.
 - This should update lifetimes.
- Stateless auto-configuration will only configure addresses.
 - It will not do all the host configuration you may want to do.
- RFC 4862 defines IPv6 Stateless Autoconfig

Router Advertisement

RA: Router Advertisement

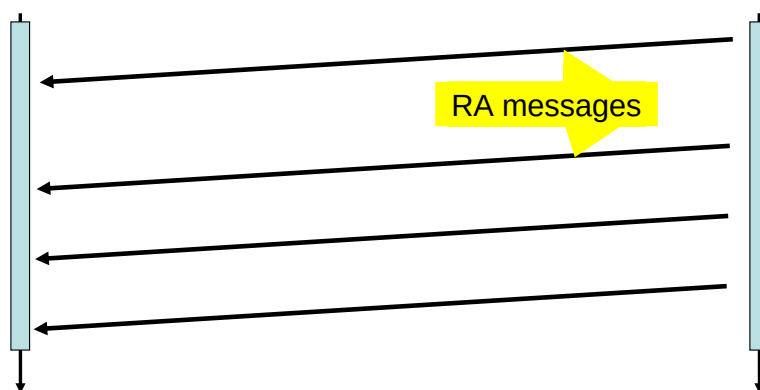
Routers periodically send advertisement messages.

Client

MAC: 00:22:cd:fc:e3:34

Router

MAC: 00:33:ad:02:e3:34

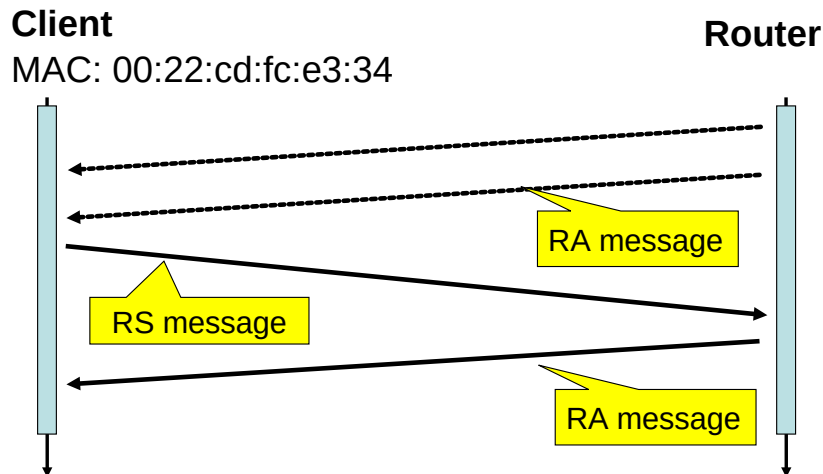


RA messages contain information about further configuration

Router Solicitation

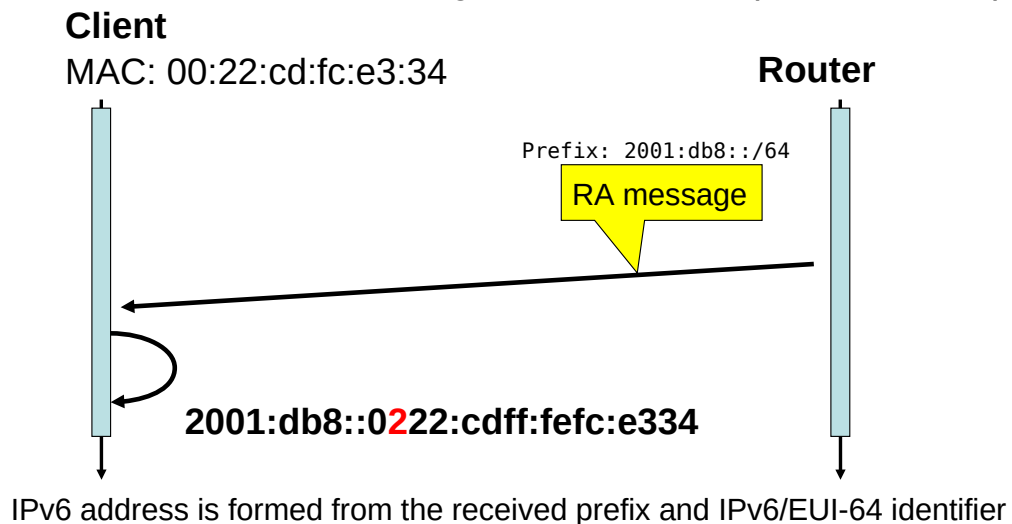
Host can wait for a periodic RA message.

Or, it can send a Router Solicitation (RS) message.



SLAAC Global Address

RA message contains the required network prefix.



This globally unique IPv6 address can then be used to communicate in the IPv6 network

SLAAC + DNS

- O mecanismo de SLAAC é orientado para configurar:
 - Endereço
 - Prefixo
 - Gateway
- Outras configurações (DNS, ...) não são implementadas.
- Soluções:
 - DHCPv6 (sem SLACC)
 - SLACC + DHCPv6
 - SLACC/IPv6 + DHCP/IPv4
 - SLACC + RDNSS Option (RFC 8106) + ...

IPv6 RDNSS Option

RA-based DNS Discovery

Autoconfiguration of **Recursive DNS Server (RDNSS)**

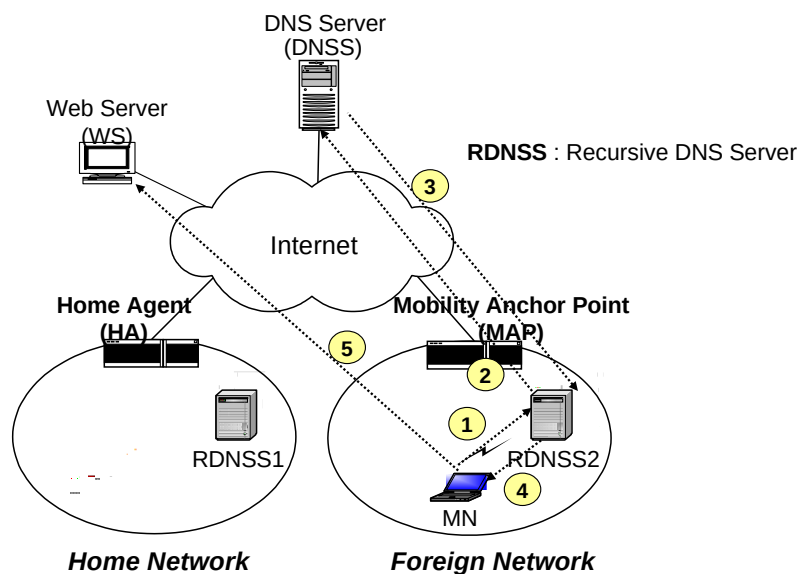
- RA message announces DNS Information to IPv6 hosts.
- An IPv6 host can select the address of RDNSS(es) received
- Not all hosts suport RDNS Option.

Optimization of DNS Name Resolution

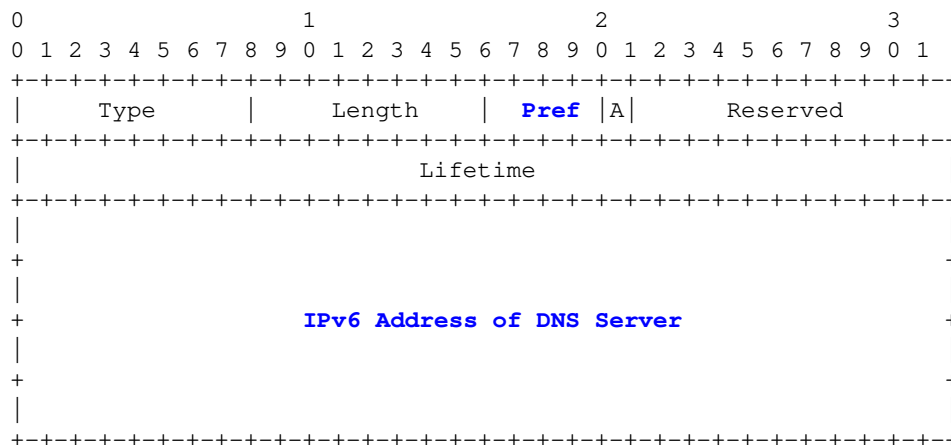
- is possible by using the local RDNSS placed in the same network.
- IPv6 Host can **select the nearest RDNSS** based on:
 - the "Pref" field in DNS Option
 - the longest prefix matching between:
 - the RDNSS's
 - Host's IPv6 addresses.

Application of DNS Autoconfiguration

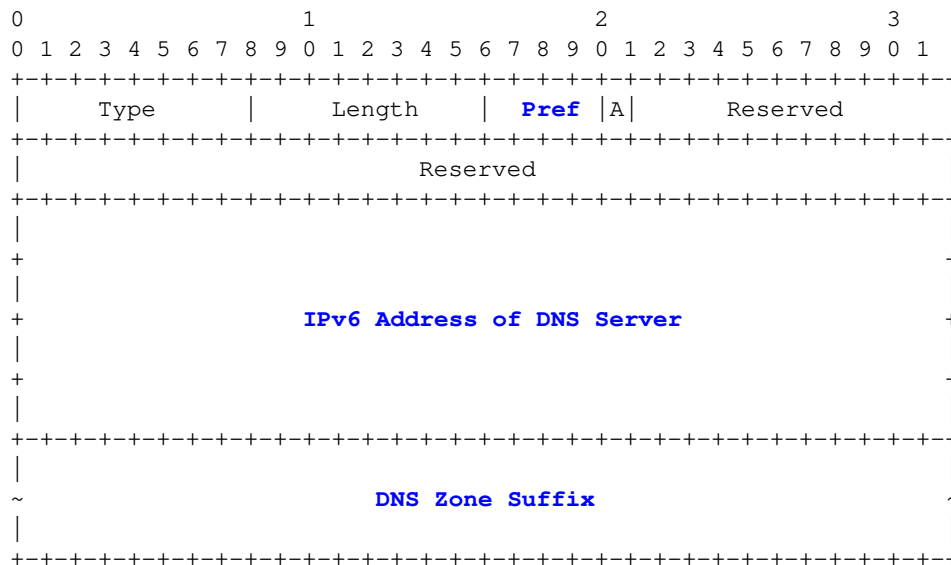
Optimization of DNS Name Resolution



RDNSS Option format



DNS Option format



IPv6 RDNSS Option

1. RA-based DNS Discovery

Autoconfiguration of **Recursive DNS Server (RDNSS)**

- RA message announces DNS Information to IPv6 hosts.
- An IPv6 host can configure the address of **RDNSS(es)** placed in the local or visited network.

Optimization of DNS Name Resolution

- is possible by using the **local RDNSS** placed in the same network.

2. Auto-Registration of Host DNS Name

A **User-preferred Host DNS Name**, including the Recursive DNS Server **DNS zone suffix** of the local or visited network, can be registered with DNS server through DNS Dynamic Update.

DNS Search List

Discovery of DNS Zone Suffix

RA based method.

Router informs the domain name suffix.

DNSL Option format

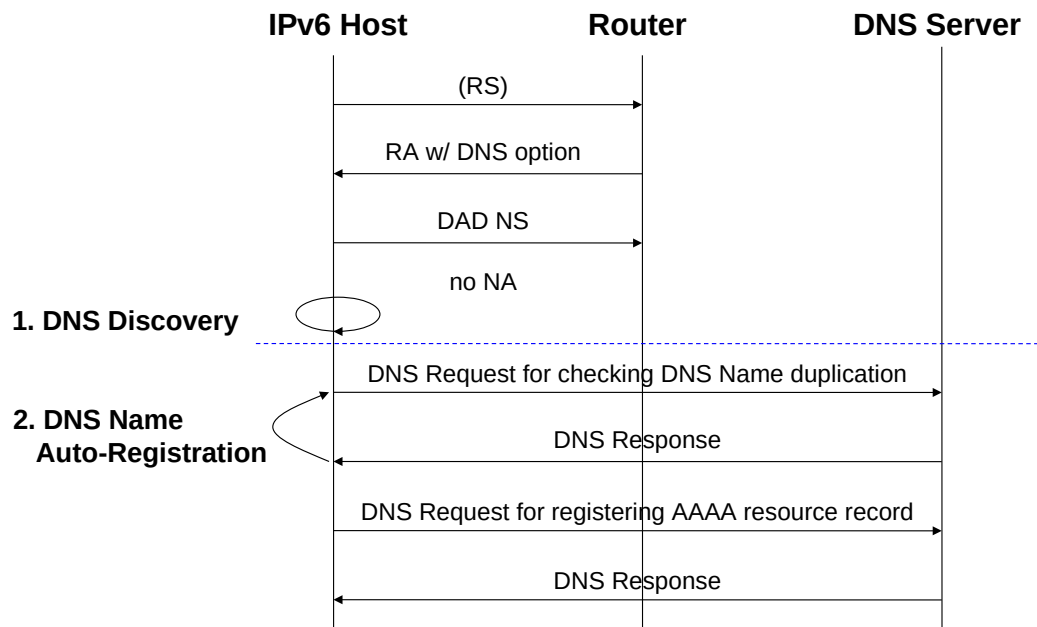
```

0                               1                               2                               3
0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|      Type      |      Length      |  Pref  |A|      Reserved      |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|                                                         |
|                               Lifetime                               |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|                                                         |
| ~      Domain Names of DNS Search List      ~ |
|                                                         |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
```

DNS Name Auto-Registration

Discovery of DNS Zone Suffix
For DNS Name Auto-Registration

Procedure of DNS Autoconfiguration



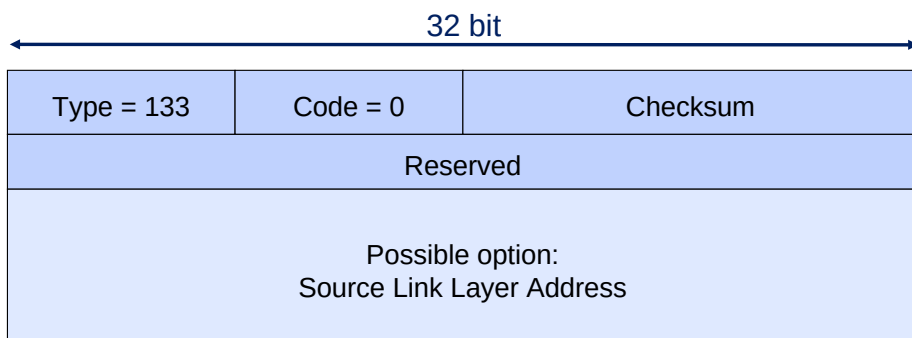
NDP

NDP: Neighbor Discovery Protocol

- ➔ Replaces ARP, RARP, IRDP, ...
- ➔ Defines five ICMPv6 packet types.
 - Type 133: **Router Solicitation**
 - Type 134: **Router Advertisement**
 - Type 135: **Neighbor Solicitation**
 - Type 136: **Neighbor Advertisement**
 - Type 137: **Redirect**

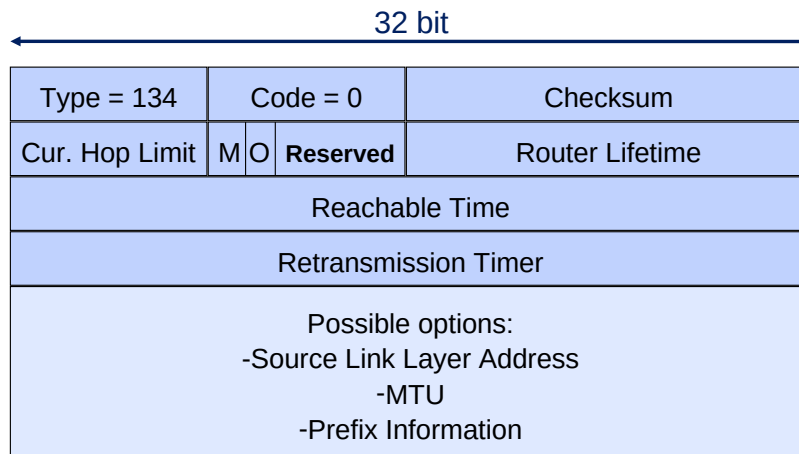
Router Solicitation

- Hosts can send Router Solicitations (RS) to ask for setup data.
- Uses ICMPv6 packet with type=133, code=0
- RS are sent to the all-routers multicast address.

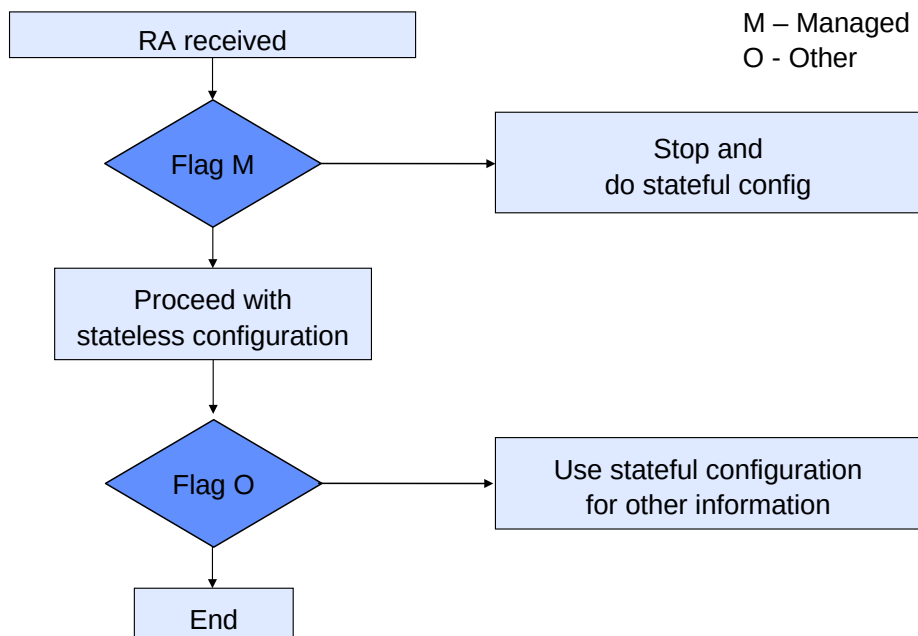


Router Advertisement

- Routers advertise their presence periodically or in response to Router Solicitations.
- RA messages can include setup info: Network prefix, MTU, ...
- Sent to the RS requester or to the all-nodes multicast address.



RA Flags



Stateful Configuration

- When stateless configuration is not desired, a configuration server (DHCP) must be provided.
- DHCP server can provide configuration information to the hosts as they come up.
 - RFC 3315 defines DHCP, updated by RFC 4361
 - Dnsmasq – DHCPv6 implementation
<http://sourceforge.net/projects/dnsmasq>

DHCPv6

DHCPv6: Dynamic Host Configuration Protocol for IPv6

- DHCP is Stateful,
- DHCP requires a server.
- Can provide DNS server identification and other info
- Three primary configuration options are available for IPv6 interfaces:
 1. Use autoconfiguration and not DHCPv6 (SLAAC).
 2. Use DHCPv6 and not autoconfiguration.
 3. Get an address with autoconfiguration and then use DHCPv6 to retrieve additional information.

SLAAC pros & cons

Pros

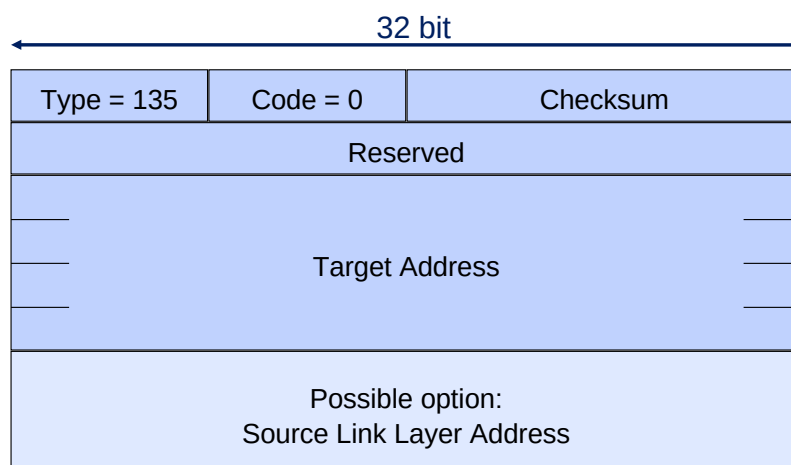
- Minimal setup
- Simplifies readdressing (ISP migration)
- Does not need a DHCPv6 server, only a router

Cons

- Removes control
- Adds vulnerabilities
- Lack ability to provide added information like:
 - DNS servers
 - NTP servers
 - Fallback gateways
- Allocates the last 64 bits of the address

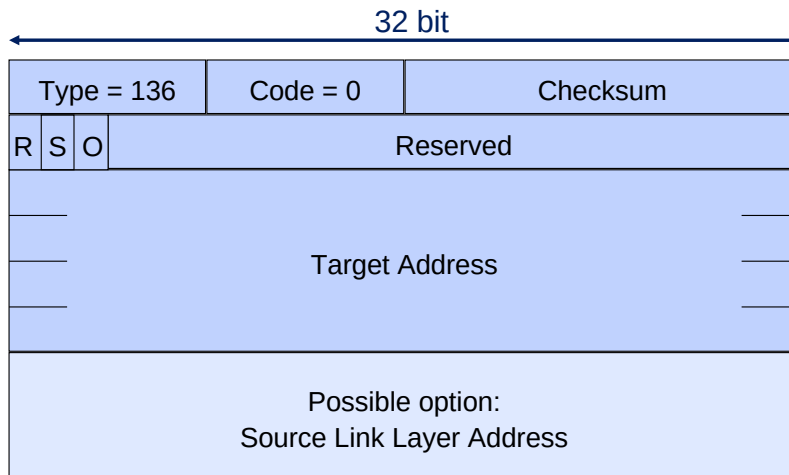
Neighbor Solicitation

- Used to request (or verify) the link layer address of a neighbor.
- Also used to DAD: Duplicate Address Detection.
- Uses ICMPv6 packet with type=135, code=0.
- Sent to the multicast address (request/DAD) or to the target (verify).

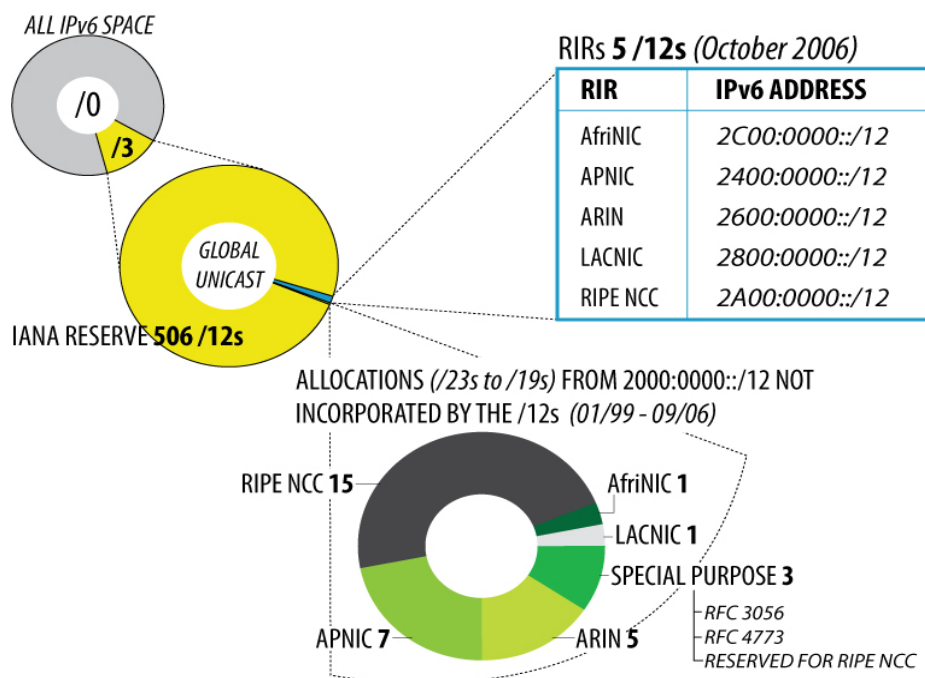


Nighbor Advertisement

- Used to inform neighbors about our link layer address.
- Uses ICMPv6 packet with type=136, code=0
- NA are sent to the NS requester or the all-nodes multicast address



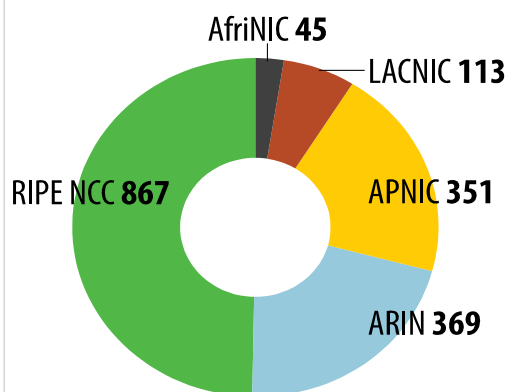
Address allocation



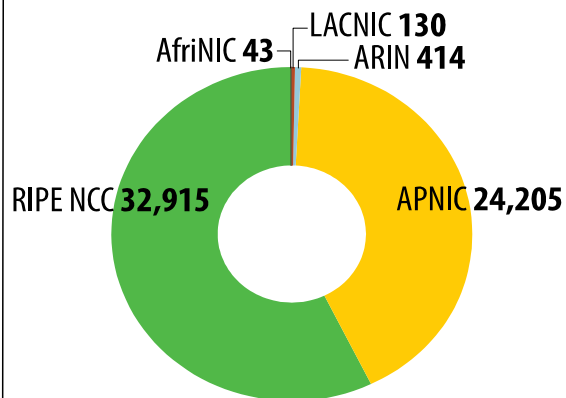
IPv6 ALLOCATIONS RIRs To LIRs/ISPs

(Jan 1999 – March 2008)

How many total allocations have been made by each RIR?



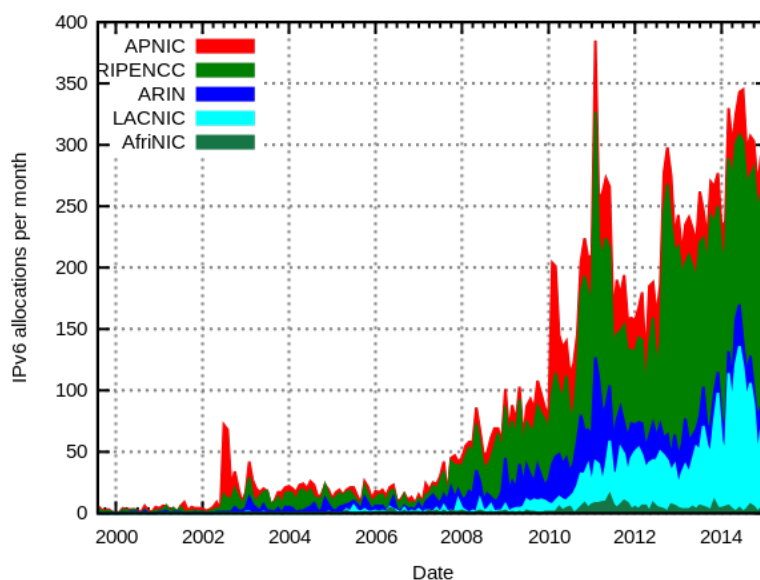
In terms of /32s, how much total space has each RIR allocated?



Dec 2007

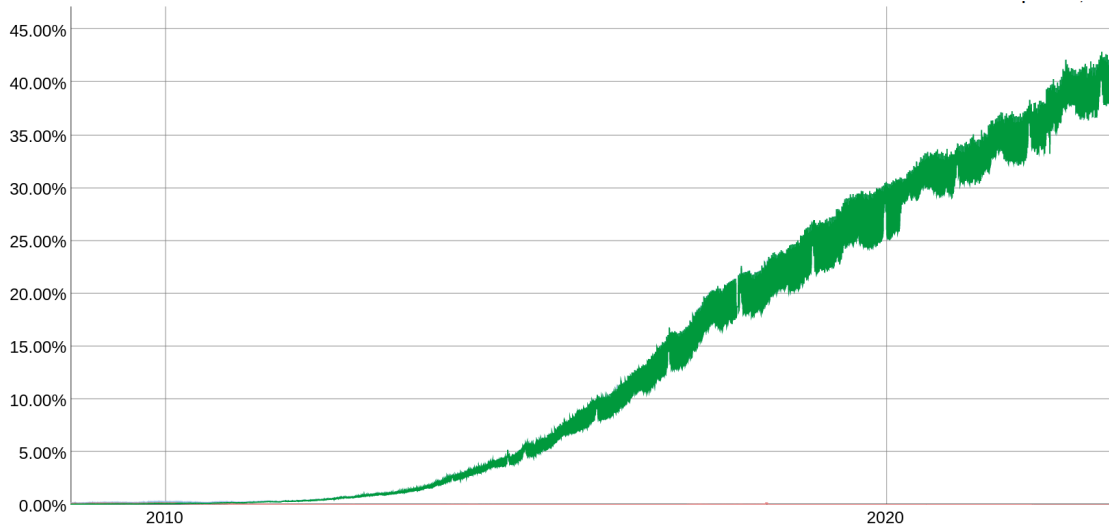
Internet Number Resource Report

Monthly IPv6 allocations per RIR

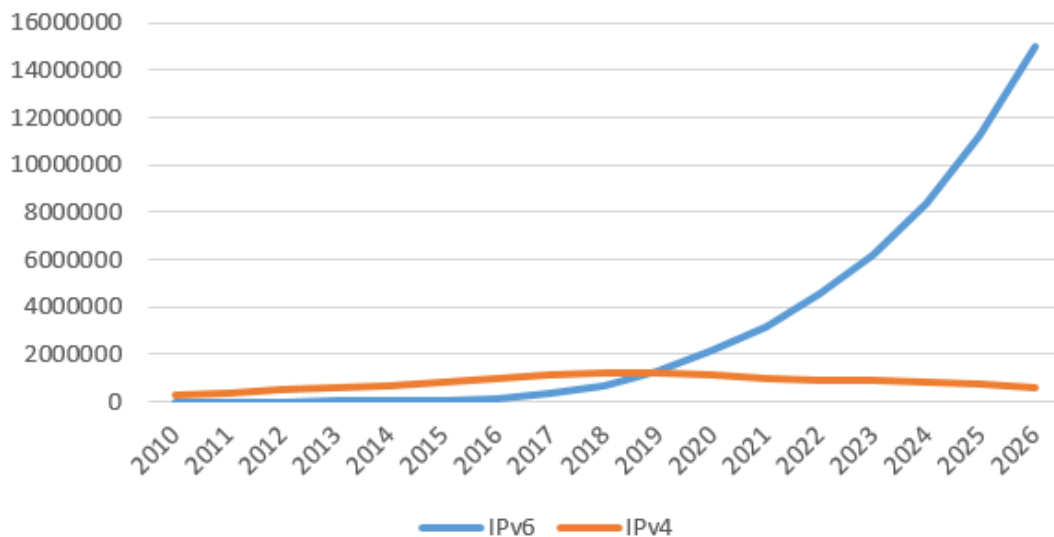


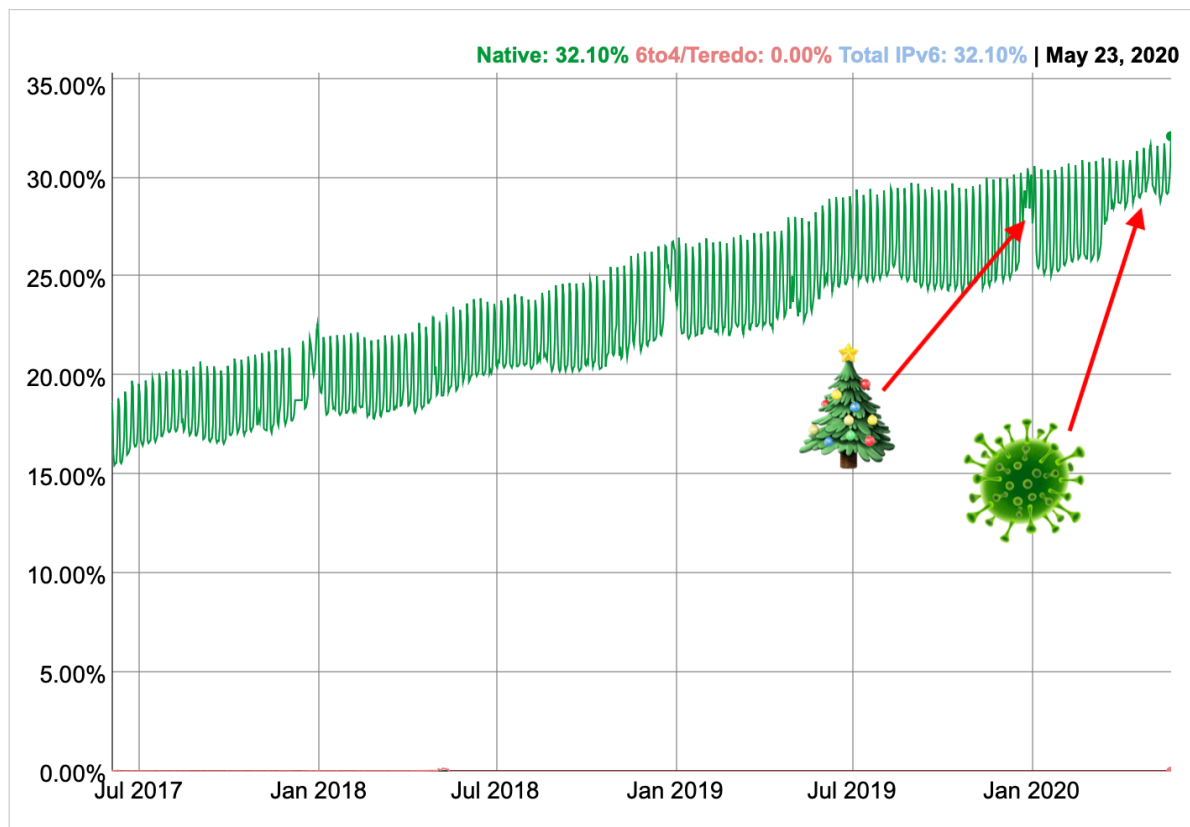
Utilização de IPv6

Google IPv6

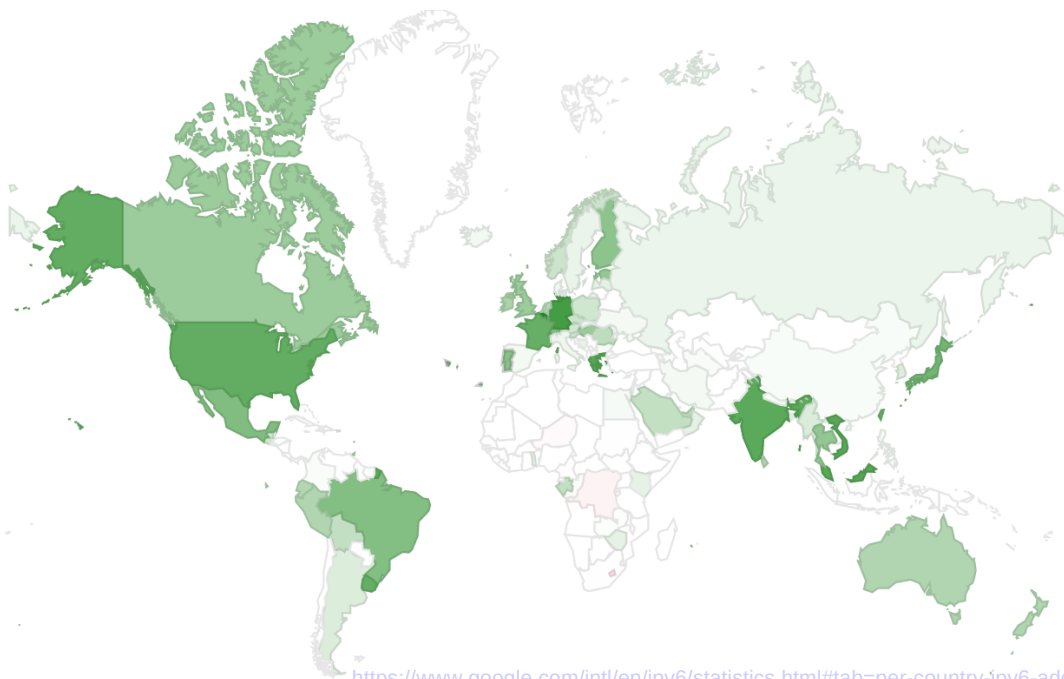


IPv6 Overtakes IPv4

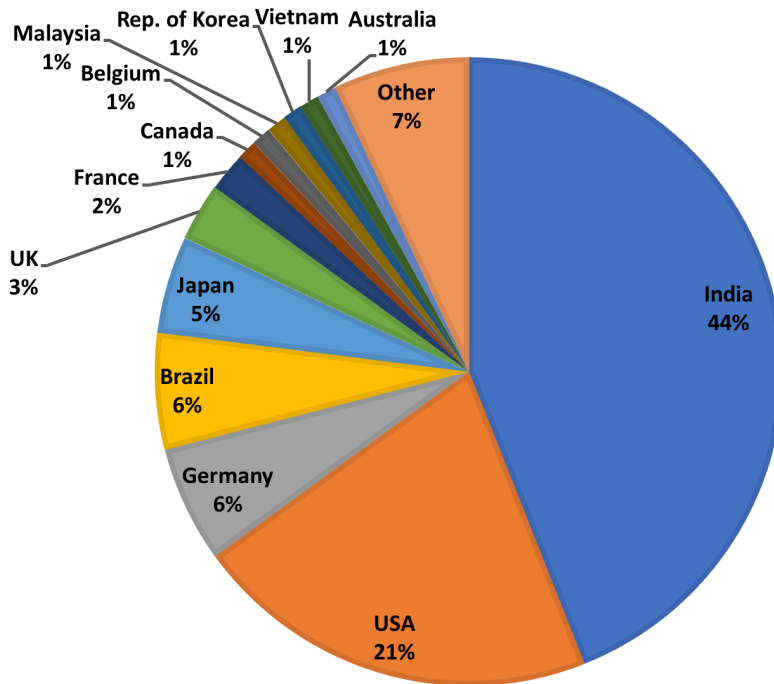




Utilização de IPv6 por país



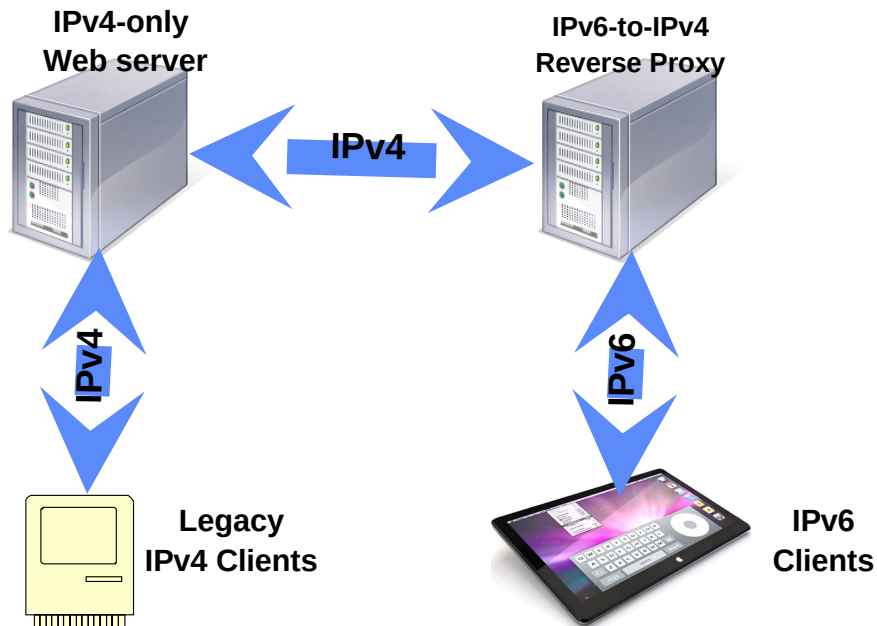
Utilizadores IPv6



Países com menos de 5% de IPv6



IPv4-to-IPv6 Reverse Proxy



Bibliografia Recomendada

- ➔ William Stallings, Data and Computer Communications, 7th Edition, Prentice Hall, Chapter 18 - Internet Protocols
- ➔ Andrew S. Tanenbaum, Computer Networks, 3rd Edition, Prentice Hall, Chapter 5.5.10

Router Solicitation Options Prefix Information

- This should include all prefixes the router is aware of
- Flag bits:
 - On-link = 1
 - Prefix is specific to the local site
 - Autonomous Configuration bit = 1
 - Use the prefix to create an autonomous address

Prefix Option

Type	Length	Prefix Length	L	A	Reserved
Valid Lifetime					
Preferred Lifetime					
Reserved					
Prefix List					

Router Solicitation Options

Prefix Information

- Valid & preferred lifetime values in router-advertisements can be used for address renumbering.
- Valid Lifetime
 - 32-bit unsigned integer. The length of time in seconds before an address is invalidated.
 - During a prefix's valid life, existing connections can be used, but new connections may not be opened.
- Preferred Lifetime
 - 32-bit unsigned integer. The length of time in seconds before an address is deprecated.
 - During a prefix's preferred life, new connections can be opened at will.

Stateless Autoconfiguration

